

LAB 1 (Please record your answers on this document!)

Live Troubleshooting Challenge

The aim of this exercise is to assess real-world network troubleshooting:
symptoms → verification → root cause → correction → validation.

CST4585 – Troubleshooting Assessment

- **Topic:** VLANs, Trunking, and DHCP Misconfiguration
 - **Assessment Type:** Hands-on Troubleshooting + Written Justification
 - **Estimated Time:** 45–60 minutes
-

Scenario Overview

A previously working enterprise access/distribution network is now experiencing **connectivity and addressing issues** affecting Staff and Student hosts.
You are **not told what was changed**.

Your task is to:

- Identify the fault(s)
 - Explain the root cause
 - Correct the configuration
 - Verify full restoration of service
-

Part 1 – Observed Symptoms

Answer the following **based on testing from the PCs and switches**.

Question 1:

PC1 (Staff VLAN) is unable to communicate beyond its local switch.

- Can PC1 obtain a DHCP address?
- Can PC1 ping its default gateway?

Record test results:

```
PC1-STAFF:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP qlen 1000
    link/ether 52:54:00:60:08:96 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::5054:ff:fe60:896/64 scope link
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN qlen 1000
    link/ether 52:54:00:50:58:d9 brd ff:ff:ff:ff:ff:ff
PC1-STAFF:~$
```

DHCP Request → Failed

Gateway Ping → Failed

External Connectivity → Failed

Question 2:

PC2 (Student VLAN) is unable to obtain a valid IP address.

- What IP address (if any) is assigned?
- Is the subnet correct for VLAN 20?

Record test results:

```
PC2-STUDENT:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP qlen 1000
    link/ether 52:54:00:96:95:9a brd ff:ff:ff:ff:ff:ff
    inet6 fe80::5054:ff:fe96:959a/64 scope link
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN qlen 1000
    link/ether 52:54:00:74:dc:27 brd ff:ff:ff:ff:ff:ff
PC2-STUDENT:~$
```

- PC2 is assigned an APIPA address (169.254.x.x) because it cannot reach the DHCP server.
- No, the subnet is incorrect. The DHCP pool is configured with the wrong network, which prevents PC2 from receiving a valid VLAN 20 IP address.

Question 3

Inter-switch connectivity appears inconsistent.

- Are all expected VLANs visible across switches?
- Do trunks appear operational?

Provide evidence using command output:

```
A1#show vlan brief
VLAN Name                Status    Ports
-----
1    default                active    Et0/0, Et0/2, Et1/0, Et1/1
20   VLAN0020                active    Et1/2, Et1/3
1002 fddi-default            act/unsup
1003 token-ring-default    act/unsup
1004 fddinet-default        act/unsup
1005 trnet-default         act/unsup
A1#show interface trunk
Port      Mode          Encapsulation  Status        Native vlan
Et0/1     on            802.1q         trunking      1

Port      Vlans allowed on trunk
Et0/1     1-4094

Port      Vlans allowed and active in management domain
Et0/1     1,20

Port      Vlans in spanning tree forwarding state and not pruned
Et0/1     20
A1#
```

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```
D1#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Et0/0, Et0/1, Et1/0, Et1/1 Et1/2, Et1/3
10 staff	active	
20 student	active	
1002 fddi-default	act/unsup	
1003 token-ring-default	act/unsup	
1004 fddinet-default	act/unsup	
1005 trnet-default	act/unsup	

```
D1#show interface trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Et0/2	on	802.1q	trunking	1
Et0/3	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Et0/2	1-4094
Et0/3	1-4094

Port	Vlans allowed and active in management domain
Et0/2	1,10,20
Et0/3	1,10,20

Port	Vlans in spanning tree forwarding state and not pruned
Et0/2	1,10,20
Et0/3	1,10

```
D1#
```

```
D2#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Et0/0, Et1/0, Et1/1, Et1/2 Et1/3
10 staff	active	
20 student	active	
1002 fddi-default	act/unsup	
1003 token-ring-default	act/unsup	
1004 fddinet-default	act/unsup	
1005 trnet-default	act/unsup	

```
D2#show interface trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Et0/1	auto	n-802.1q	trunking	1
Et0/2	auto	n-802.1q	trunking	1
Et0/3	auto	n-802.1q	trunking	1

Port	Vlans allowed on trunk
Et0/1	1-4094
Et0/2	1-4094
Et0/3	1-4094

Port	Vlans allowed and active in management domain
Et0/1	1,10,20
Et0/2	1,10,20
Et0/3	1,10,20

Port	Vlans in spanning tree forwarding state and not pruned
Et0/1	1,10,20
Et0/2	1,10,20
Et0/3	20

```
D2#
```

- No, all expected VLANs are not visible across the switches because trunk configuration is incorrect or missing, preventing VLAN information from being shared between switches.
- No, trunks do not appear fully operational because the trunk interface is configured as an access port instead of trunk mode, which blocks VLAN traffic between switches.

Part 2 – Fault Isolation (Guided Troubleshooting)

Fault Area A – Access Port VLAN Assignment

Task A1

Verify VLAN membership on the access switch.

```
show vlan brief
```

Question A1:

Which interface is incorrectly assigned to the wrong VLAN?

- Interface: Ethernet0/2
 - Current VLAN: Vlan 1
 - Expected VLAN: Vlan 10
-

Fault Area B – Trunk Configuration

Task B1

Verify trunk status on the access switch.

```
show interfaces trunk
```

Question B1:

Is Ethernet0/0 operating as a trunk?

- Current mode: Access mode
- Expected mode: Trunk Mode

Question B2:

Explain how this misconfiguration impacts VLAN traffic between switches.

Ans :When trunking is disabled, VLAN traffic cannot pass between switches. This prevents communication between VLANs, blocks DHCP requests from reaching the server, and causes connectivity failures for devices in different VLANs.

Fault Area C – DHCP Configuration

Task C1

Inspect the DHCP pool configuration on the distribution switch.

```
show ip dhcp pool
```

Question C1:

What subnet is configured for STUDENT_VLAN20_POOL?

```
D1#show ip dhcp pool

Pool STAFF_VLAN20_POOL :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 30
  Leased addresses                : 0
  Excluded addresses             : 6
  Pending event                  : none
  1 subnet is currently in the pool :
  Current index    IP address range    Leased/Excluded/Total
  10.0.0.1         10.0.0.1 - 10.0.0.30    0 / 6 / 30

Pool STUDENT_VLAN20_POOL :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 30
  Leased addresses                : 0
  Excluded addresses             : 0
  Pending event                  : none
  1 subnet is currently in the pool :
  Current index    IP address range    Leased/Excluded/Total
  10.0.1.1         10.0.1.1 - 10.0.1.30    0 / 0 / 30
D1#
```

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The pool is using 10.0.1.0/24, but VLAN 20 should be 10.0.0.32/27. So the DHCP pool is wrong.

Question C2:

Why does this subnet prevent Student clients from receiving valid addresses?

```
00#show ip dhcp pool
Pool STAFF_VLAN20_POOL :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 30
  Leased addresses : 0
  Excluded addresses : 0
  Pending event : none
  1 subnet is currently in the pool :
  Current index   IP address range      Leased/Excluded/Total
  10.0.0.1        10.0.0.1 - 10.0.0.30  0 / 0 / 30

Pool STUDENT_VLAN20_POOL :
  Utilization mark (high/low) : 100 / 0
  Subnet size (first/next) : 0 / 0
  Total addresses : 30
  Leased addresses : 0
  Excluded addresses : 0
  Pending event : none
  1 subnet is currently in the pool :
  Current index   IP address range      Leased/Excluded/Total
  10.0.1.1        10.0.1.1 - 10.0.1.30  0 / 0 / 30

00#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/          Lease expiration    Type      State      Interfa-
ce               ce
-----
Hardware address/
User name
```

Because the DHCP pool is using the wrong network, the server gives out IPs that don't match VLAN 20. So PC2 gets no valid IP

Part 3 – Root Cause Analysis (Written)

For each fault, complete the table:

Fault	Root Cause	Impact
Incorrect VLAN on access port	Access port assigned to VLAN 1 instead of VLAN 10	Staff PC cannot reach DHCP server or gateway
Trunking disabled	Inter-switch link configured as access instead of trunk	VLAN traffic cannot pass between switches
Incorrect DHCP network	DHCP pool configured with wrong subnet	Student devices receive invalid or no IP address

Part 4 – Corrective Actions (Hands-On)

Students must **apply fixes**, not just describe them.

Fix 1 – Correct Access VLAN

```
interface Ethernet0/2
 switchport access vlan 10
```

Fix 2 – Restore Trunking

```
interface Ethernet0/0
 switchport mode trunk
```

Fix 3 – Correct DHCP Pool

```
ip dhcp pool STUDENT_VLAN20_POOL
 network 10.0.0.32 255.255.255.224
```

Part 5 – Verification (Mandatory)

Students must **prove resolution**.

Required Verification Commands

show vlan brief

```
A1#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Et1/0, Et1/1, Et1/2, Et1/3
10	VLAN0010	active	Et0/2
20	VLAN0020	active	Et0/3
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

```
A1#show interfaces trunk
```

Vlan 10 and Vlan 20 are active

show interfaces trunk

```
A1#show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Et0/0	on	802.1q	trunking	1
Et0/1	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Et0/0	10,20
Et0/1	1-4094

Port	Vlans allowed and active in management domain
Et0/0	10,20
Et0/1	1,10,20

Port	Vlans in spanning tree forwarding state and not pruned
Et0/0	10
Et0/1	1,20

Ethernet0/1 is now in trunk mode

show ip dhcp binding

There is no binding because the PCs cannot run due to permission error

End-Device Testing

- PC1 receives correct VLAN 10 address
- PC2 receives correct VLAN 20 address
- Both PCs can ping their default gateways
- PC1 can ping PC2

Part 6 – Reflection Questions (Assessment Depth)

1. Why does placing an access port in VLAN 1 often cause *silent failures*?

Ans : VLAN 1 is the default VLAN and is usually not configured for user traffic. Devices connected to VLAN 1 may appear connected but cannot access required network services.

2. Why is trunking critical even if only one VLAN appears to be in use?

Ans : Trunking allows VLAN information to pass between switches. Without trunking, VLAN communication fails and network segmentation breaks.

3. How could DHCP misconfiguration be mistaken for a Layer 2 issue?

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Ans: DHCP depends on VLAN connectivity. If VLAN or trunk configuration is incorrect, DHCP requests fail, making it appear as a Layer 2 network issue.
